

Investigating Implicit Adaptivity for an LLM-based Companion for low-acuity Patients in the Emergency Department

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Abstract

In emergency departments (ED), patients' experiences often involve needs beyond medical care, including empathetic communication while feeling seen and informed. My thesis aims to address a care gap caused by staffing constraints and high patient numbers: advancements in artificial intelligence (AI) offer an opportunity to support low-acuity patients during passive waiting times through an *AI companion* that prepares for consultation, provides information on ED processes, and offers empathetic support without replacing clinical staff. Based on workshops and patient interviews, preliminary requirements for this companion concept were gathered. Building on these insights, my next steps focus on investigating how the companion can *implicitly* adapt its role and behavior to individual patients, without the cognitive burden of explicitly customizing it. My thesis contributes to the growing body of literature on patient-facing conversational AI and explores how adaptive systems can enhance patient experiences while supporting clinical workflows in high-stress ED settings.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

AI, LLMs, co-design, patients, emergency department, companion

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1 Background and Motivation

Due to the care gap caused by rising patient numbers and limited staff availability [10], emergency departments (EDs) worldwide are facing systemic challenges. Clinical staff face increased workloads [6], while patients are experiencing a decline in the quality of care [17]. Low-acuity walk-in patients, who often require less urgent medical treatment [26], face particular vulnerabilities beyond their immediate medical needs. These include prolonged waiting times [4, 9, 17, 36], limited access to information [19] and compassionate care [9], communication barriers, often due to medical

jargon [8, 39, 52], and feelings of not being taken seriously [19, 39]. During waiting periods, they often assume a passive role while clinical staff need to prioritize more urgent cases [9, 46, 52], a state further exacerbated by fear, pain, and uncertainty [8, 39, 52].

Current capabilities of artificial intelligence (AI) and large language models (LLMs) offer a new opportunity to address patients' informational and emotional needs [13]. Through natural, adaptive conversational interaction [30, 48, 51], these systems might complement rather than displace human connection in the ED while counteracting the sense of powerlessness that often characterizes patient experiences in ED waiting areas [9, 45, 52]. Prior work in human-computer interaction (HCI) in healthcare suggested that LLM capabilities can support personalized, adaptive interaction and empathetic communication [1, 51]. Conversational agents have been shown to provide meaningful informational and emotional support, and increase patient engagement [3, 27, 43, 50]. In contexts where in-person support is limited, patient-facing companions have demonstrated potential to empower patients during rehabilitation processes [18]. Similarly, conversational agents used prior to clinical consultations have been found to support patient-physician communication by collecting patient information in advance and increasing patient satisfaction [27, 28]. In line, a patient-facing chatbot was designed in collaboration with general practitioners, aiming for better preparation for consultation and improved shared decision-making [44].

Building on this body of work, my research explores how a patient-facing AI companion could support low-acuity patients during waiting times, complementing clinical workflows in the ED (see Figure 1 for a schematic description). Specifically, I investigate the design of an AI companion whose goals are:

- G1:** providing emotional and empathetic support when clinical staff are unavailable,
- G2:** explaining ED processes to make patients feel informed and reduce feelings of uncertainty,
- G3:** supporting orientation around hospital grounds,
- G4:** helping patients recall relevant medical details, and
- G5:** preparing both patients and clinical staff for consultations by gathering structured patient information.

The companion's overarching objective is to help patients feel more informed and prepared, potentially improving both their waiting experience and treatment by supporting clinical staff through better-prepared consultations. However, the diversity of the ED patient population presents a design challenge. While low-acuity patients often share similar experiential difficulties, in-situ interviews revealed that their expectations of and comfort with such an AI companion vary substantially. In particular, patients expressed divergent preferences regarding the companion's look, feel, and



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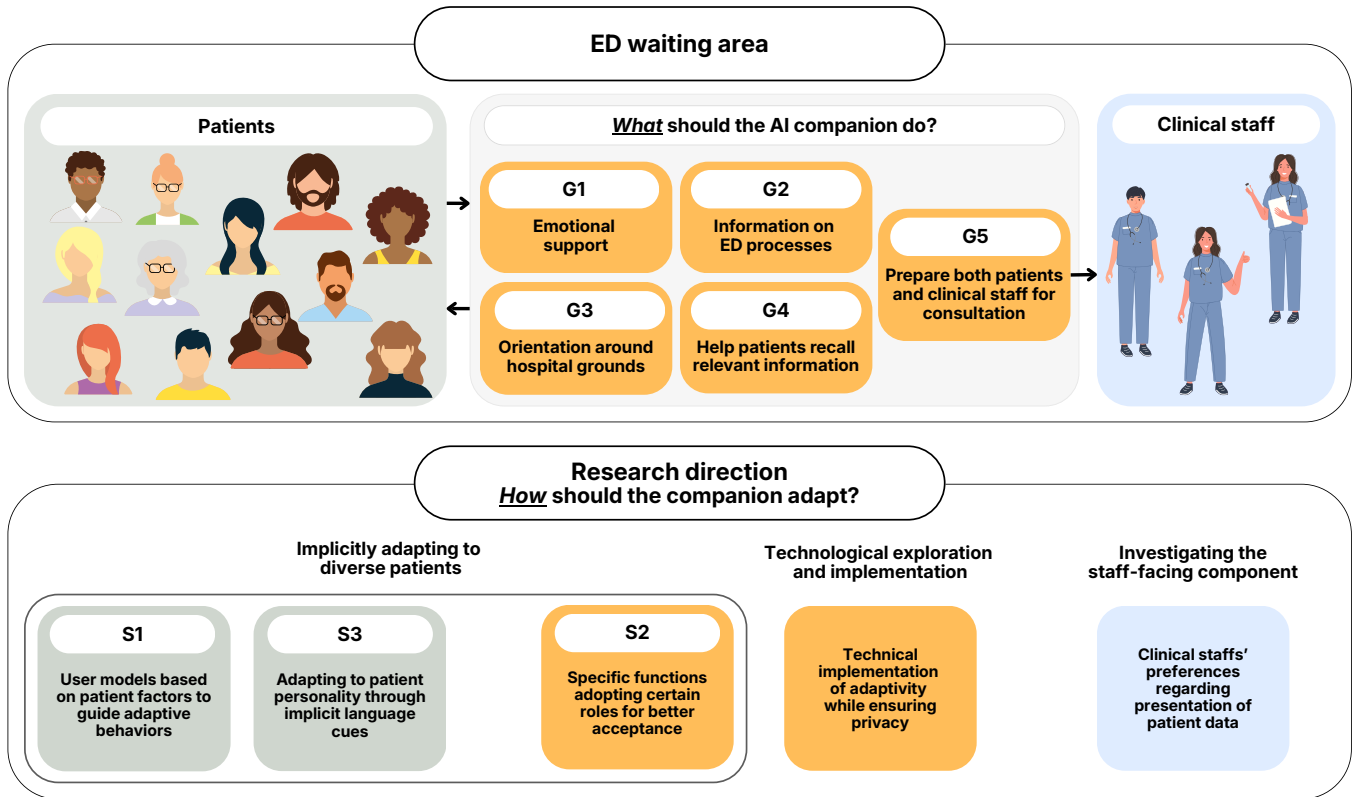


Figure 1: My research aims to design an adaptive ED AI companion that can provide emotional and informational support during times when clinical staff are unavailable and can prepare both patients and clinical staff for consultation. The diverse population of ED patients requires research on the adaptive capabilities of AI and LLMs, described in Section 3 as part of my research direction. Illustrations: © mw-images and icons via <https://canva.com>

form of interaction, suggesting that a “one-size-fits-all” design may not be appropriate, aligning with prior research [29, 38].

Consequently, my dissertation focuses on investigating how an LLM-based ED companion can adapt to individual patient needs while accounting for the cognitive and physical realities of the ED environment. Patients often experience stress, pain, and uncertainty during waiting times [8, 39, 52], which limits their ability or willingness to explicitly articulate how they want an AI system to behave. At the same time, the open space of ED waiting areas raises concerns about interactional privacy, as sharing sensitive symptoms or medical history audibly may be undesirable. Taken together, these constraints motivate my research on adaptive conversational behavior and interaction modalities that are inferred *implicitly*, aligning with the broader goal of HCI to design systems that adapt to users rather than requiring users to adapt to technology [54].

2 Contributions to date

In the first phase of my research, I focused on understanding the perspectives of both patients and clinical staff in the ED, and aimed to balance both stakeholders’ needs in the ED. Given that the ED comprises multiple stakeholders with sometimes shared needs but often divergent perspectives, I aimed to design an AI concept that addresses all needs equally, without favoring one stakeholder group

over the other. Through patient workshops, thorough discussions within our clinical research team and interviews with in-situ patients, I examined how an AI companion could be integrated in the ED waiting area without prioritizing efficiency at the expense of the patient experience. This work resulted in a first-authored paper accepted at CHI ’26 that describes a four-staged design process for deriving requirements for an ephemeral AI companion concept used during waiting times [22]. A key finding from this work is that while clinical staff often envision AI as an early point of contact to improve ED efficiency, patients strongly value initial human contact. These insights informed a design concept in which the AI companion supports patients only during waiting times, focusing on emotional reassurance, provision of information, and preparation for consultation rather than clinical decision-making.

3 Research direction

The design process outlined above, and described in detail in the full paper [22], established a shared design vision and clarified *what* the AI ED companion should do. My current and future work focuses on *how* such a companion could implicitly adapt to different patients, how this adaptivity could be realized from a technical perspective, and how the clinician-facing component of such a companion should be designed.

3.1 Implicitly adapting to diverse patients

My formative work suggests that while low-acuity patients often share similar needs regarding their overall ED experience, their perceptions of and expectations toward AI vary significantly. Building on these insights, the next phase of my research investigates how the AI companion could *adapt* to individual patients to enable truly personalized interactions [25]. Although the companion's overarching goals, i.e., offering emotional support, providing information on the ED, and preparing for consultation, are well defined, the manner in which these functions are delivered is likely to differ considerably between patients.

Rather than asking patients directly how they want the companion to behave, which might be cognitively demanding and inappropriate in the ED context [14, 37], I aim to explore how adaptivity can be inferred *implicitly*, especially in cold-start scenarios where no prior user data exists to guide such adaptivity [41]. Based on prior literature and my formative findings, I am currently exploring multiple adaptation strategies:

- S1:** Drawing on role-based models of conversational agents [42] as well as research on personality dimensions in LLM-based systems [24], I am planning an interview study with former ED patients to understand what constitutes meaningful and supportive design interaction from their perspective [40], and how they envision the companion's role. Given the diversity of patients, it is challenging to adapt the companion to each individual. Therefore, I aim to find out if we can identify different *user models* [32] that the companion could adapt to. My interview study will help determine if factors such as education [5], health literacy [23], experienced symptoms [47], and attitudes towards technology [11] can contribute to developing a heuristic user model. This model could guide adaptive interaction strategies, allowing the companion to be tailored in different roles and personalities depending on these models [32].
- S2:** Beyond a heuristic model of patients, I am also interested in whether specific functions of the companion are perceived as more acceptable or effective when associated with particular roles or personality traits. Unlike the user model approach (S1), which tailors the companion to different patient groups, this investigation focused on whether the companion should have fixed roles and personality traits for each function. This approach suggests a dynamic behavior of the companion that adjusts based on the functionality being provided rather than the individual patient.
- S3:** The third strategy conceptualizes adaptivity as *individual-level* and comparatively stable over time. In contrast to S1, which adapts the companion based on heuristic patient models derived from contextual and background factors, this approach focuses on adapting the companion's interaction style to the patient's *personality* [2, 49]. Rather than assigning patients to predefined user groups, personality traits could be inferred implicitly from ongoing interaction cues such as language use, pacing, or expressiveness [33, 35, 49]. This research direction builds on prior findings that users tend to prefer and trust conversational agents whose behavior aligns with their own personality characteristics [15, 33, 34].

In my dissertation, I plan to investigate the trade-off between these strategies and to understand whether one strategy, or a combination thereof, is more appropriate in the demanding context of the ED.

3.2 Technological exploration and implementation

Beyond the conceptual adaptive strategies, I plan to explore technological approaches that could support the companion's goals. Candidate methods include chain-of-thought reasoning [41, 53], retrieval-augmented generation [55], and reinforcement learning [31]. More specifically, I aim to use these different technological approaches to probe how implicit adaptivity can be achieved.

In parallel, patient feedback highlighted the need for privacy-preserving interaction modalities beyond digital data protection: typing is found cumbersome by some, and verbal disclosure in the open waiting area is uncomfortable. I am therefore investigating whether approaches such as *Silent Speech Interfaces* [7, 16, 20] could be further researched to facilitate private communication.

To validate these explorations, I plan to implement a prototype with a locally running LLM within the hospital environment, ensuring that sensitive data remains secure and GDPR-compliant¹. This prototype will be validated with in-situ patients using a combination of qualitative interviews and user interaction logs, providing data to refine adaptive behavior, interaction modalities, and the overall effectiveness and desirability of the companion.

3.3 Investigating the staff-facing component

Lastly, to position the companion as a meaningful and useful part in the clinical workflow, I also plan to extend the companion with a staff-facing component that summarizes patient-provided information prior to consultation [27], building a bridge between patients and clinical staff. By transferring structured, patient-generated data to clinical staff, the companion may reduce documentation burden [12], prepare clinical staff for consultation, and support more attentive, empathetic patient-physician interactions [21]. I intend to involve clinical staff in co-creation workshops to understand their needs regarding information presentation and further requirements for the companion. This holistic approach aims to ensure that the system supports both patient well-being and clinical practice.

4 Conclusion

My dissertation focuses on the design and implementation of an adaptive, patient-facing AI companion that supports low-acuity ED patients during their waiting times. By integrating perspectives from HCI, AI, and healthcare, my work seeks to design AI systems that offer emotional support, are privacy-conscious, and responsive to diverse patient needs while remaining aligned with clinical workflows. It contributes empirical insights into the realities of the clinical environment and design considerations for adaptive conversational agents. The outcomes of this research aim to inform the design of patient-facing AI systems not only in the ED but also similar systems in other critical situations where expert support is limited but emotional and informational needs are high.

¹GDPR stands for the General Data Protection Regulation in the European Union. See <https://gdpr-info.eu/>.

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